BROOKBANKS

1335 Swindon Nev

Technical Note TN47- OVL/A420 Signalised Junction S278 Surface Water Design
11th January 2023

1 Executive Summary

1.1 Swindon Eastern Villages Consortium have appointed Brookbanks to produce a surface water drainage design for a residential-led scheme development at South Marston and Rowborough to the east of Swindon, together with areas of mixed-use comprising community; employment; education; local retail uses; and formal and informal open space provision on land east of Swindon and north of A420.

1.2 The scheme is to include an upgrade to the existing junction between the A420 and Old Vicarage Lane (OVL) to be adopted under a S278 agreement. The junction will require an approved drainage system to discharge Condition 27– Old Vicarage Lane/A420 (ref. S/OUT/13/1555).

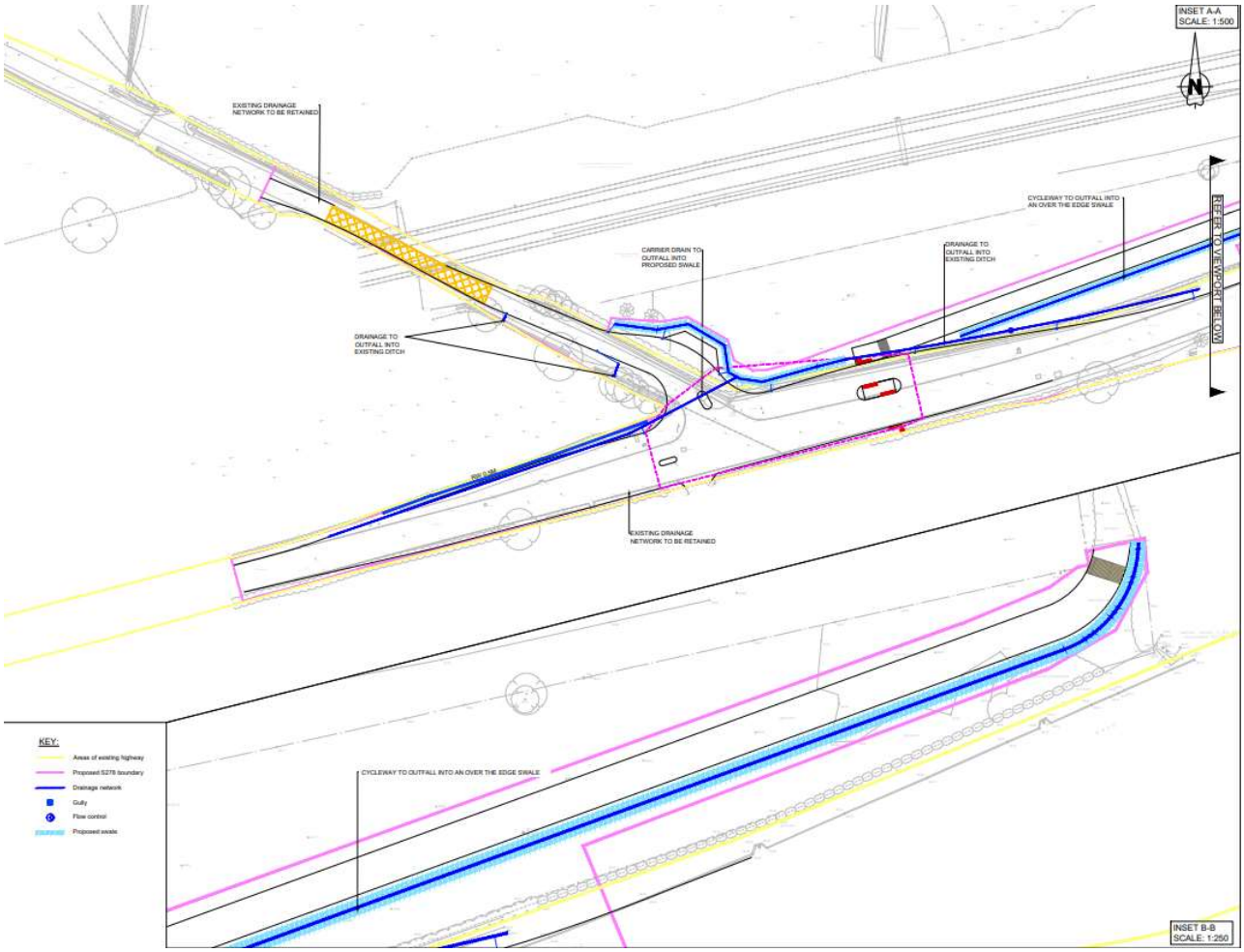
1.3 We believe this can be achieved with the provision of additional gullies and carrier drains along the A420 in combination with a swale to attenuate and treat the surface water run off as set out in this note and on drainage plan ref **1335-HL-058-500D Drainage Layout**. The proposal accords with the approved Flood Risk Assessment 1335 FRA01 Rv3 and the reports prepared to discharge Conditions 40 & 41 relating to site wide drainage and flood mitigation.

2 Proposed Junction

2.1 The proposed layout consists of

- An upgraded signalised junction on the A420
- A segregated cycle link from the site to the southern edge of the S278 works
- A maintenance layby
- A pedestrian and cycle crossing

Shown in the extract below.



Proposed drainage layout (Extract Brookbanks GA ref 1335 -HL-58-500 D)

3 Surface Water Run Off

Existing Highway Drainage System

3.1 A topographical survey has been undertaken to verify the extent of the existing highway drainage system draining the OVL/A420. Brookbanks have visited site for a walkover and observed the existing highway drainage system to consist of a combination of over the edge drainage and gully inlets to an existing ditch on either side of the A420.

3.2 The proposal will be to retain and improve the existing drainage system where the alignment remains unchanged.

Proposed Highway SuDS

3.3 The S278 works will result in an increase of surface water runoff. In line with approved FRA and SWDS a new highway SuDS system is proposed to treat and attenuate runoff as illustrated in **Drawing 1335-HL-058-500 Drainage Layout**.

3.4 The proposal will include

- a swale to intercept the runoff from the proposed cycleway.
- a swale to the east of Old Vicarage Lane
- Additional gullies and carrier drains where the widening of the junction will not allow over the edge drainage.
- Outfall to the existing ditchcourse

A typical swale arrangement will be 3.4m wide approx. 0.5m deep with 1 in 3 side slopes.

The primary aims of the swale is as follows:

- Final flow and water quality conditioning
- Provide landscaping, amenity and ecological benefits.

Outfall

3.5 Flows will be restricted by a control device and will outfall to the existing ditch on the northside of the A420 Clearance work will be undertaken to the existing ditch course to ensure flow is maintained.

Supporting measurements and calculations

3.6 We have calculated the following using Microdrainage Source Control Software for storm events up to and including a 1 in 100-year event plus 40% climate change.

- Additional Impermeable area approximately **0.2ha**
- Flow restricted to **5l/s** to mimic existing flows from the site.
- Volume of storage required approximately **120m^3**

3.7 The proposed highway drainage arrangements, including the incorporation of swales, have resulted in no increase to the flow rates to the existing highway system or surrounding drainage network up to and including the 1 in 100-year storm event plus 40% climate change.

4 Maintenance Plan

Highway Drains

4.1 Once constructed and when the adoption process has been completed, Swindon Borough Council will be responsible for maintaining the highway drainage networks and ensuring their continued operation. Consequently drains, manhole and headwalls will be maintainable by the SBC.

SuDS

4.2 The swales are to be maintained by a Swindon Borough Council in accordance with a Maintenance & Management Plan to be approved by the LPA/LLFA. Detailed plans and design drawings indicating features and areas which will be maintained by Swindon Borough Council will be provided in the Maintenance Plan.

4.3 The development layout provides adequate access and working areas to undertake the above maintenance procedures.

4.4 The Council maintenance team will maintain the basin in accordance with best practice based on a typical maintenance regime shown in the tables below:

Maintenance Schedule	Required Action	Typical Frequency
Regular Maintenance	Remove litter (including leaf litter) and debris	Monthly or as required
	Cut grass for spillways and access routes	Monthly during growing season or as required
	Manage other vegetation and remove nuisance plants	Monthly at start, then as required

	Inspect inlets, outlets and overflows for blockages, and clear if required.	Monthly
	Inspect banksides, structures, pipework etc for evidence of physical damage	Monthly or when required
	Inspect inlets and facility surface for silt accumulation. Establish appropriate silt removal frequencies	Monthly for 12 months, then annually or as required
	Inspect inlets and facility surface for silt accumulation, and establish appropriate silt removal frequencies	Six Monthly
	Check any penstocks and other mechanical devices	Annually
	Tidy all dead growth before start of growing season	Annually
	Remove sediment from inlets, outlet and forebay	Annually or as required
	Manage wetland plants in outlet pool- where provided	Annually
Occasional Maintenance	Reseed areas of poor vegetation growth.	As required
	Prune and trim any trees and remove cuttings	Every 2 years or as required
	Remove sediment from inlets, outlet and forebay and main basin	Every 5 years or as required
Remedial Actions	Repair erosion or other damage by re-turfing or reseeding	As required
	Re-alignment of riprap	As required
	Repair/ rehabilitate inlets, outlets and overflows	As required
	Relevel uneven surfaces and reinstate design levels.	As required

Exceedance Flows

4.5 The wide verge and swale will allow a route for exceedance flows ensuring any flooding in extreme events is diverted away from habitable dwellings.

5 Next Steps and Recommendations

5.1 The proposed swale system is designed in accordance with the approved FRA to allow deliver of the proposed S278 signalised junction.

5.2 The developer to ensure that the contractor constructs and maintains the drainage system until the system is adopted by Swindon Borough Council